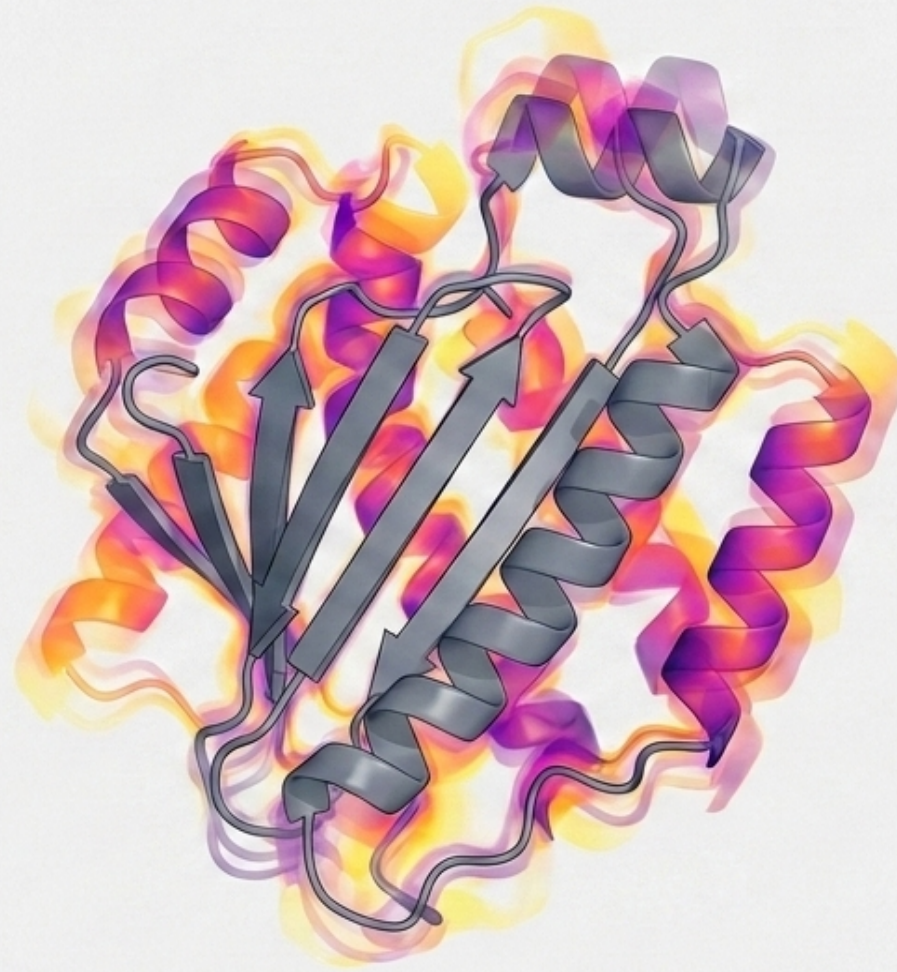




ENSEMBITS: an alphabet of protein conformational ensembles

Kaiwen Shi, Carlos Oliver

ENSEMBITS_01

Motivation

From Static Structures to Conformational Dynamics

Dynamics as the Language of Function: Biological function is often defined not by a single static 3D "snapshot," but by the distribution of conformations a protein samples and the correlated motions connecting them.

Example: G-protein coupled receptors (GPCRs) must transition between active and inactive states to relay signals

The Representation Bottleneck: Existing protein structure tokenizers (PSTs) only capture static, local geometry. They miss the "correlated motions and alternative conformational states" that define an ensemble.

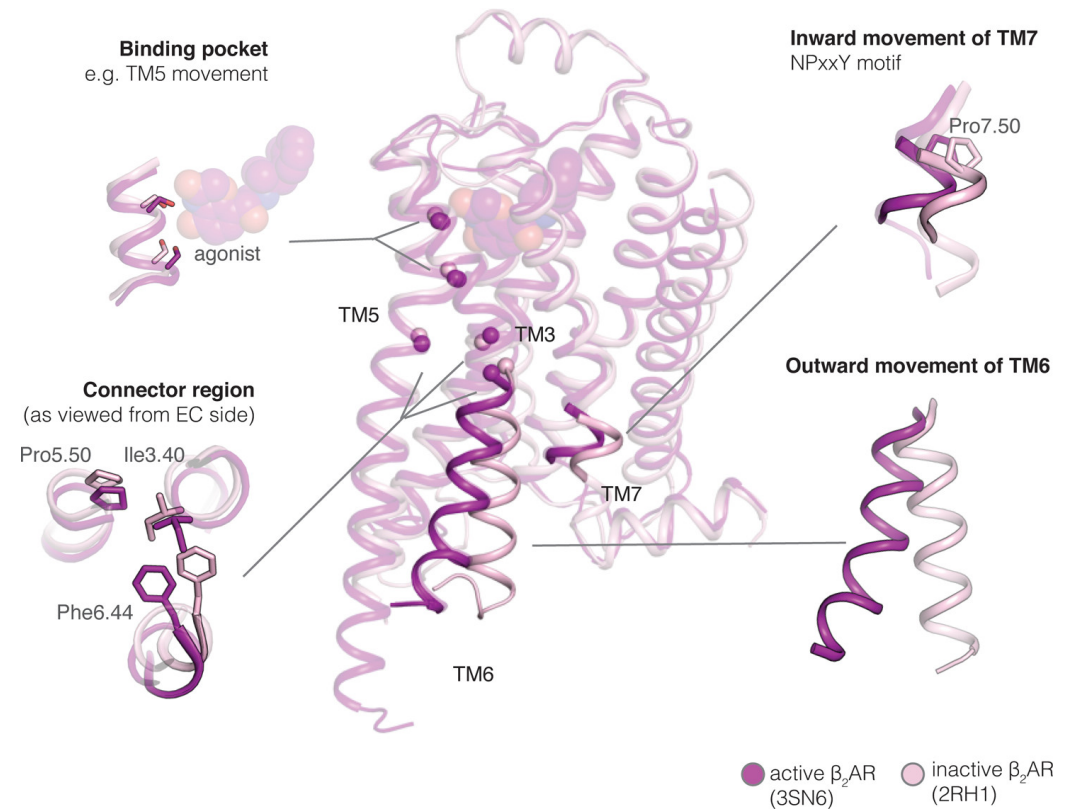


Figure 6. Latorraca, N. R., Venkatakrishnan, A. J., & Dror, R. O. (2017). GPCR Dynamics: Structures in Motion. *Chemical Reviews*, 117(1), 139–155. <https://doi.org/10.1021/acs.chemrev.6b00177>

Motivation

From Static Structures to Conformational Dynamics

Data Availability: Large-scale molecular dynamics (MD) repositories and new generative models (like AlphaFlow or BioEmu) are providing an vast amount of ensemble data compared to prior bottleneck. However this data is not yet processed into a format Protein Language Models (PLMs) can use.

Dynamics Alphabet: There are already structural alphabets like 3Di. Prior work like ProtProfileMD, and "Learning dynamic protein representations at scale with distograms" depend on histograms/distograms of static tokens, which cannot model how different residues move in a correlated fashion.

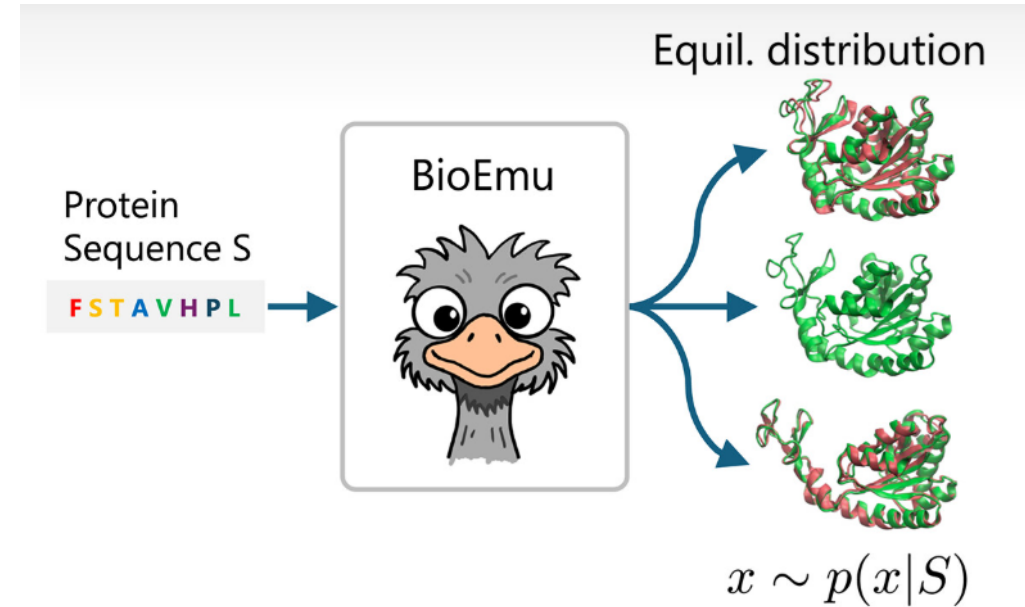


Figure 1 (Cropped). Lewis, S., Hempel, T., Jiménez-Luna, J., Gastegger, M., Xie, Y., Foong, A. Y. K., Satorras, V. G., Abdin, O., Veeling, B. S., Zaporozhets, I., Chen, Y., Yang, S., Foster, A. E., Schneuing, A., Nigam, J., Barbero, F., Stimper, V., Campbell, A., Yim, J., ... Noé, F. (2025). Scalable emulation of protein equilibrium ensembles with generative deep learning. *Science*, 389(6761), eadv9817. <https://doi.org/10.1126/science.adv9817>

Method

ENSEMBITS

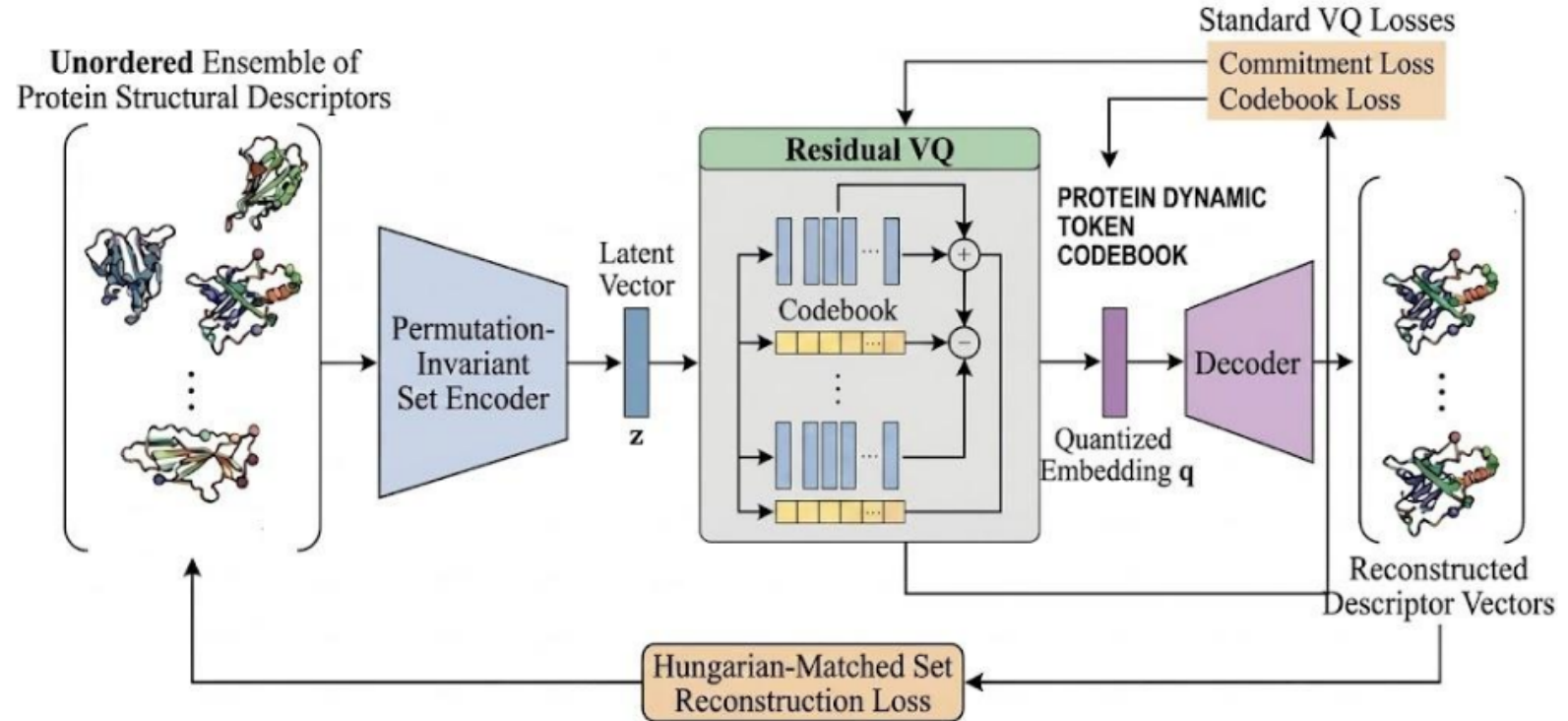


Figure 1: The Ensembits tokenization pipeline. The P -frame per-residue descriptor multiset is encoded by a permutation-invariant set encoder, residually quantized into a token tuple $(C_1, \dots, C_M)$ with token embedding q , and decoded back to P descriptors under a Hungarian-matched reconstruction loss. Codebooks are updated by EMA; the alternative gradient-based codebook loss is shown for completeness.

Method

Problem Formulation

$P = \{\mathbf{x}^1, \dots, \mathbf{x}^P\}$: an unordered multiset of P frames of a single protein

$\mathbf{x}^p \in \mathbb{R}^{L \times d}$: the (per-atom) coordinates of the L residues in conformation p

$d = 3A$: per residue coordinate dimensions, A is the number of atoms represented per residue (1 for only C_α)

An SE(3)-invariant **descriptor** maps **a frame and a residue index** to a per-residue, per-frame descriptor $\mathbf{f}_r^p$ of dimension D_f .

$$\phi(\mathbf{x}^p, r) = \mathbf{f}_r^p$$

A **dynamics tokenizer** sends the per-residue descriptor multiset

$\{\mathbf{f}_r^1, \dots, \mathbf{f}_r^P\}$ to a token in a **codebook** $\mathcal{C}$ of M entries. Each token C_m

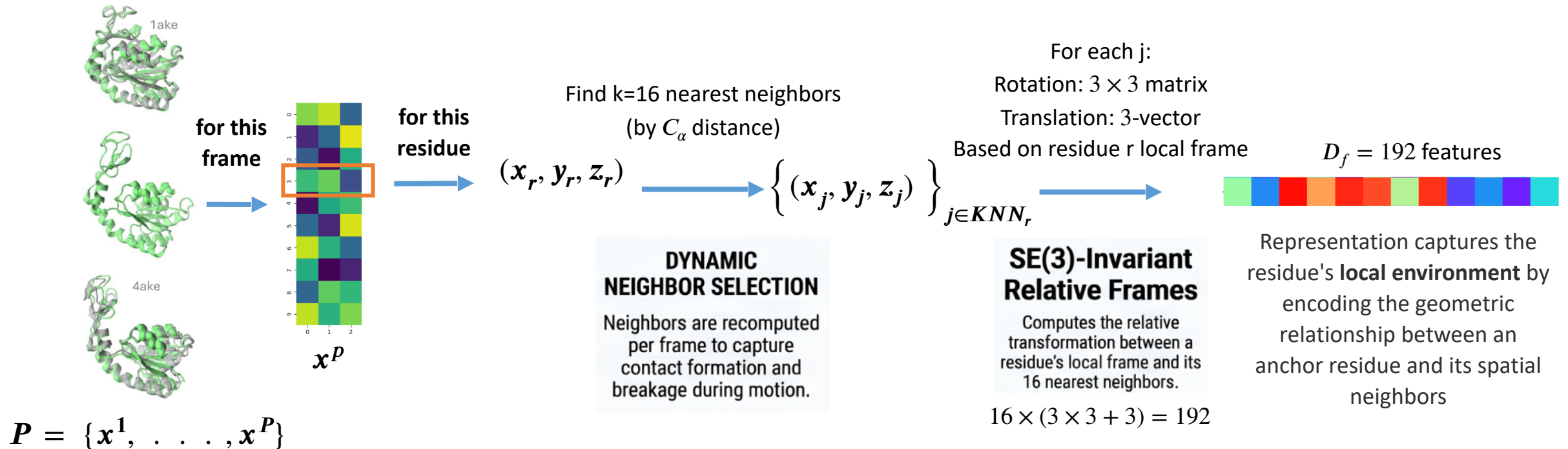
$$T : \bigcup_{P \geq 1} (\mathbb{R}^{D_f})^P \longrightarrow \mathcal{C} = \{C_1, \dots, C_M\}$$

carries an **embedding** $\mathbf{q}_{C_m} \in \mathbb{R}^D$.

p: frame / conformation, **r**: residue index, **L**: residue count, **C**: codebook, **M** : codebook count ,
 $\mathbf{x}^p$: coordinates of all residues in conformation p

Method

Descriptor: ESM3 style relative-frame descriptor



Adenylate kinase

p: frame / conformation, r: residue index, x^p : coordinates of all residues in conformation p

Method

Descriptor: 3Di style descriptor

$$i: \text{current residue}, j: \text{neighbor} \quad \begin{aligned} u_{a \rightarrow b} &= \text{unit}(C_{\alpha_b} - C_{\alpha_a}) & dir_r &= \text{unit}(C_{\alpha_{r-1}} \rightarrow C_{\alpha_{r+1}}) \\ & & cd &= \text{unit}(C_{\alpha_{j_m}} \rightarrow C_{\alpha_{j_{m+1}}}) \end{aligned}$$

#	Feature	Definition
0	d_{ij}	$\ C\alpha_i - C\alpha_j\ $
1	a_1	$\langle u_{i-1 \rightarrow i}, u_{i \rightarrow i+1} \rangle$ (bend at i)
2	a_2	$\langle u_{j-1 \rightarrow j}, u_{j \rightarrow j+1} \rangle$ (bend at j)
3	a_3	$\langle u_{i-1 \rightarrow i}, u_{i \rightarrow j} \rangle$
4	a_4	$\langle u_{j-1 \rightarrow j}, u_{i \rightarrow j} \rangle$
5	a_5	$\langle u_{i-1 \rightarrow i}, u_{j \rightarrow j+1} \rangle$
6	a_6	$\langle u_{i \rightarrow i+1}, u_{j-1 \rightarrow j} \rangle$
7	a_7	$\langle u_{i-1 \rightarrow i}, u_{j-1 \rightarrow j} \rangle$ (backbone alignment)
8	seq_1	$\text{sign}(i - j) \cdot \min(i - j , 4)$
9	seq_2	$\text{sign}(i - j) \cdot \log(i - j + 1)$

Per neighbor 3Di features (10)

$$\begin{aligned} \sin\psi_i \\ \cos\psi_i \\ \sin\psi_j \\ \cos\psi_j \end{aligned}$$

Backbone
dihedrals (4)

#	Feature	Definition
0	Distance	$\ C\alpha_{j_m} - C\alpha_{j_{m+1}}\ $
1	Alignment	$\langle \text{dir}_{j_m}, \text{dir}_{j_{m+1}} \rangle$
2	Approach	$\langle \text{dir}_{j_m}, cd \rangle$
3	Twist	$\langle \text{dir}_{j_{m+1}}, cd \rangle$

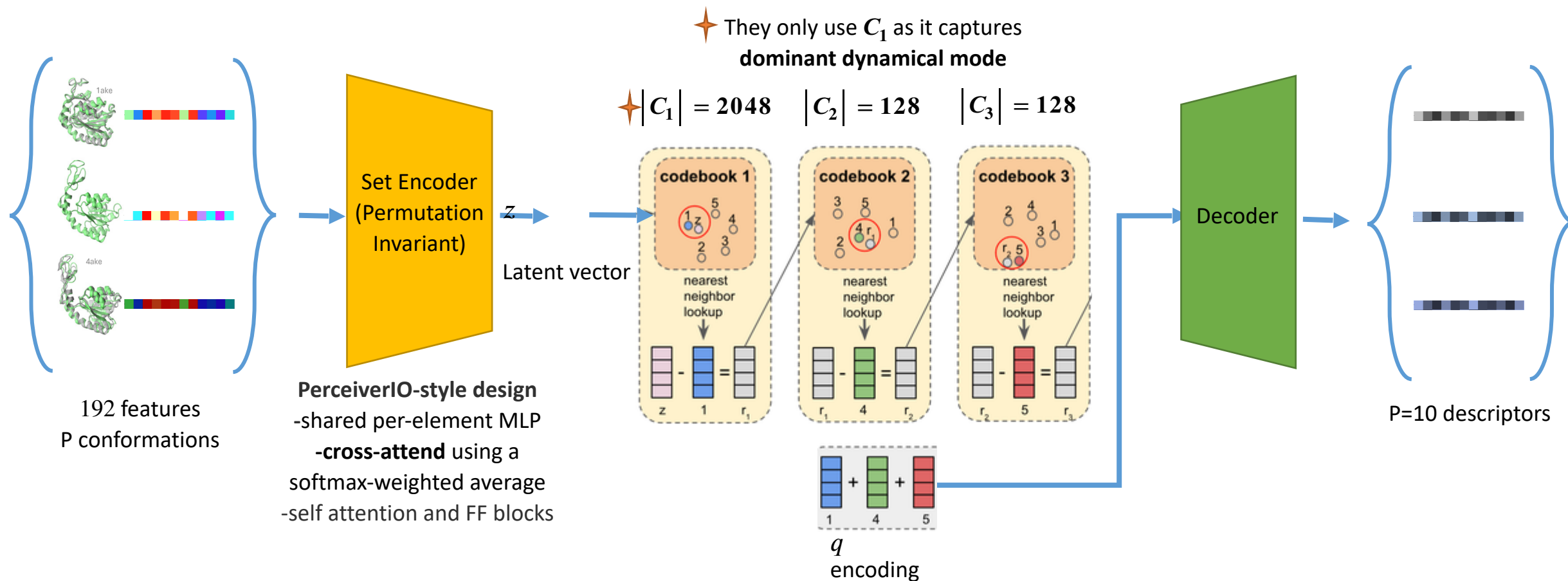
Inter-Neighbor Glue
Features (GeoBPE) (4)

Per-residue layout. Let n be the number of neighbour slots per residue. The descriptor is the concatenation

$$[3\text{Di}_1, \text{dih}_1, \text{glue}_{1 \rightarrow 2}, 3\text{Di}_2, \text{dih}_2, \text{glue}_{2 \rightarrow 3}, \dots, 3\text{Di}_n, \text{dih}_n],$$

Method

Tokenizer: ESM3 style relative-frame descriptor



Method

Single Frame to Token Distillation

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{commit}} + \lambda \cdot \|z_2 - \text{sg}[z_1]\|_2^2,$$

z_1 : Set encoder latent variable for the entire frame set

z_2 : Set encoder latent variable for a randomly sampled subset of the frame set

$\mathcal{L}_{\text{recon}}$: Hungarian-matched set reconstruction loss

$\mathcal{L}_{\text{commit}}$: Standard VQ commitment loss

Datasets

Training Datasets

The production model was trained on a **combined corpus** of approximately 19,414 protein ensembles.

- **mdCATH-div**: A **homology-deduplicated** derivative of the mdCATH all-atom molecular dynamics (MD) corpus. It contains approximately 2,442 representative domains.
 - Homology deduplication: splits selected from the H-superfamily level of the CATH classification to prevent homology leakage during evaluation
 - frames from the 320K trajectory
 - Farthest-Point Sampling (FPS) to select maximally diverse subsets of size 3, 5, 10
- **MISATO**: An MD dataset consisting of approximately 16,972 protein–ligand complexes derived from PDBbind. Each entry includes a 10 ns simulation in explicit water
 - FPS to select maximally diverse 8- and 10-frame ensembles

Datasets

Validation and Functional Benchmarking Datasets

ProteinShake: Used to create the **sequence, structure, and random-clustered splits** for the MISATO downstream suite of functional tasks

- **Sequence Split:** In this configuration, proteins are clustered by UniProt identity so they do not share significant similarity, though they may still share a common structural class.
- **Structure Split:** This is the most challenging setting where splits don't share neither sequence identity nor structural class.
- **Random Split:** This is the easiest setting used as a sanity check, where proteins are drawn at random for training and testing without any specific constraints on their similarity.

PROTEINGYM: The substitution benchmark was used for **zero-shot mutational fitness prediction**, specifically restricted to 96 deep-mutational-scanning assays with wild-type sequences of 200 residues or less.

AFsample2 was used to generate synthetic ensembles to test a data augmented version of the model (**ENSEMBITS_aug**).

Experiments

Inspecting the Tokens: Stationary Example

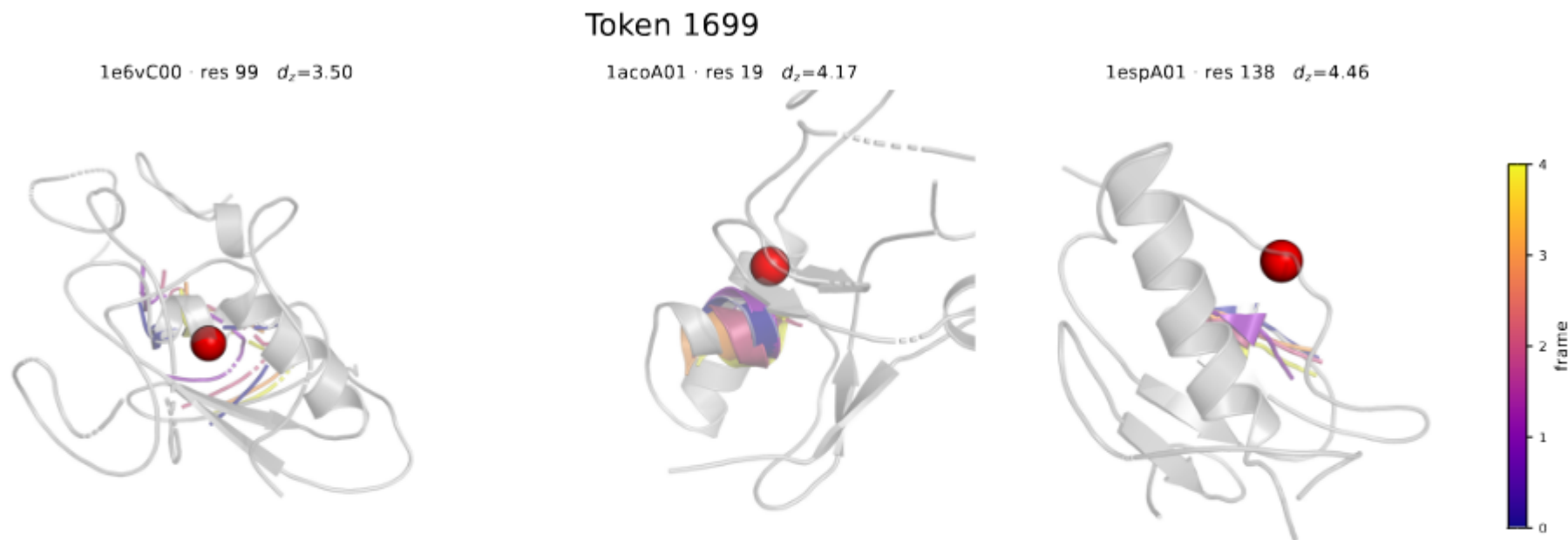


Figure 2: Token 1699 — a near-stationary local motif. Three distinct-protein exemplars (1e6vC00:99, 1acoA01:19, 3DCT:26), ranked left-to-right by encoder-latent distance d_z to the codebook centroid. Each panel overlays five MD frames Kabsch-aligned on all C_α atoms with the highlighted regions excluded. The frames superimpose tightly both within each panel and across exemplars, with mean motion amplitude, the first PCA singular value, $\langle s_1 \rangle \approx 2.2$.

Experiments

Inspecting the Tokens: Flexible Example

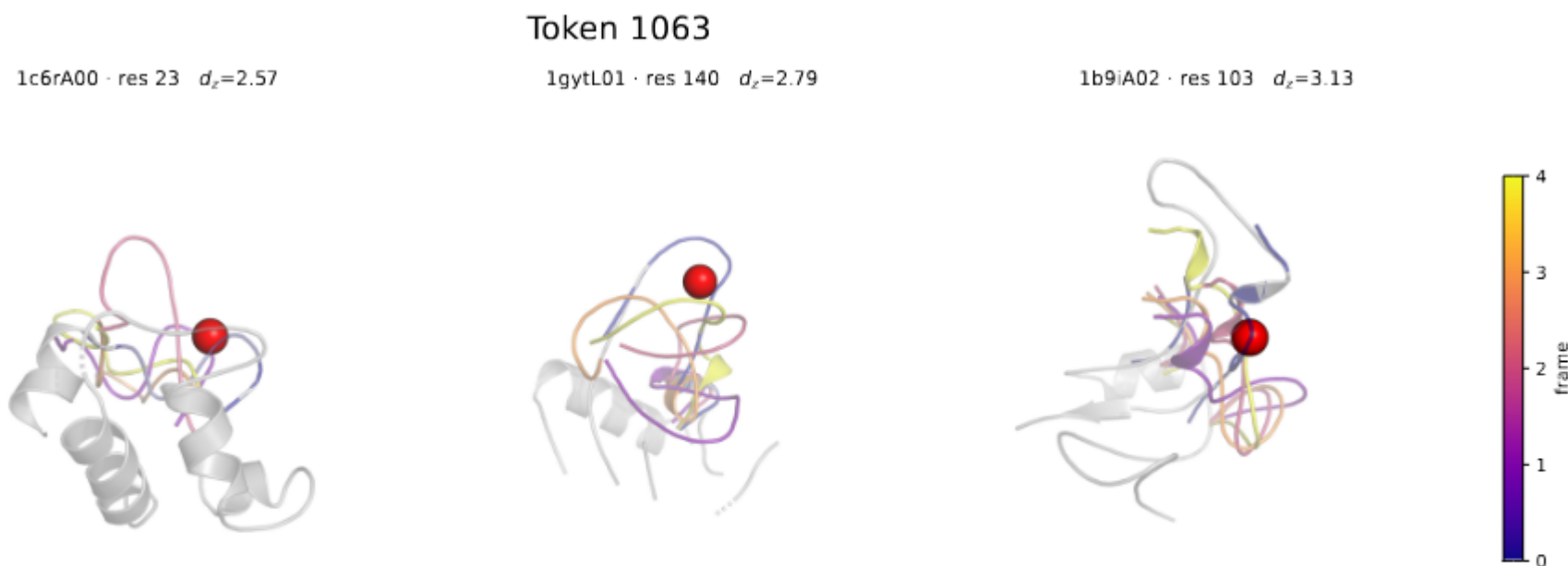


Figure 3: Token 1063 — a flexible local motif. Three exemplars (1c6rA00:23, 1gytL01:140, 1b9iA02:103) ordered left-to-right by increasing d_z (2.57, 2.79, 3.13). Same alignment and coloring as Fig. 2; mean motion amplitude $\langle s_1 \rangle \approx 8.9$, roughly four times higher than Token 1699.

Experiments

Flexibility: RMSF Prediction

Table 1: RMSF Spearman across two datasets, mean $\pm$ std over 10 seeds. mdCATH-div uses the CATH-H structure split with $P = 10$ frames; MISATO is reported on both the sequence (UniProt-clustered) and structure (CATH-H clustered) splits with $P = 8$ frames. The best tokenizer is **bold**, and the second best underlined.

Model	mdCATH-div	MISATO (seq)	MISATO (struct)
AA	0.125 $\pm$ 0.000	0.212 $\pm$ 0.001	0.235 $\pm$ 0.002
3Di_tokens	0.337 $\pm$ 0.000	0.360 $\pm$ 0.000	0.302 $\pm$ 0.001
Random	0.009 $\pm$ 0.002	0.006 $\pm$ 0.001	0.004 $\pm$ 0.001
AminoAseed	<u>0.408 $\pm$ 0.001</u>	<u>0.500 $\pm$ 0.000</u>	<u>0.488 $\pm$ 0.002</u>
ProToken	0.031 $\pm$ 0.001	0.059 $\pm$ 0.001	0.024 $\pm$ 0.002
ESM3struct	0.365 $\pm$ 0.001	0.461 $\pm$ 0.000	0.411 $\pm$ 0.001
ENSEMBITS, $P = 1$	0.458 $\pm$ 0.000	0.513 $\pm$ 0.000	0.531 $\pm$ 0.002
ProtProfileMD_K	0.579 $\pm$ 0.001	0.420 $\pm$ 0.001	0.367 $\pm$ 0.001
Vote_3Di	0.429 $\pm$ 0.000	0.366 $\pm$ 0.000	0.323 $\pm$ 0.001
ENSEMBITS _aug, $P = \text{full}$	<u>0.592 $\pm$ 0.000</u>	<u>0.523 $\pm$ 0.000</u>	<u>0.545 $\pm$ 0.001</u>
ENSEMBITS, $P = \text{full}$	0.604 $\pm$ 0.001	0.532 $\pm$ 0.001	0.550 $\pm$ 0.001

Experiments

Flexibility: ANOVA test C_α fluctuation amplitude

s_1 : Local **motion amplitude**, calculated by taking a 10-frame trajectory of a residue, applying Kabsch-alignment to remove global rigid-body motion, and computing the **top singular value** (from SVD) of the resulting coordinate matrix

- s_1 small -> rigid
- s_1 large -> flexible

Variance Decomposition (ANOVA):

$$\underbrace{\sum_r (s_1(r) - \bar{s}_1)^2}_{SS_{\text{total}}} = \underbrace{\sum_t n_t (\bar{s}_1^{(t)} - \bar{s}_1)^2}_{SS_{\text{between}}} + \underbrace{\sum_t \sum_{r \in t} (s_1(r) - \bar{s}_1^{(t)})^2}_{SS_{\text{within}}},$$

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}} \in [0, 1].$$

- $\eta^2 = 0$ -> no variance explained by tokens
- $\eta^2 = 1$ -> all variance explained by tokens

Experiments

Flexibility: ANOVA test C_α fluctuation amplitude

s_1 : Local motion amplitude, η^2 : proportion of total variance explained by the token

Feature	η^2 (observed)	F -test	permutation null mean
motion amplitude s_1	0.371	$F(851, 301, 127) = 208.5, p < 10^{-300}$	0.0028
motion shape s_1/s_2	0.040	$F(851, 301, 127) = 14.5, p < 10^{-300}$	—

Table 3: One-way ANOVA on per-residue motion features grouped by primary token, $n = 301,979$ residues, $M = 852$ tokens with ≥ 80 residues each.

Grouping variable	# groups M	η^2 on s_1
ENSEMBITS	852	0.371
CATH structural class	5	0.054
Protein-length quintile	5	0.054
Residue position quintile (within chain)	5	0.020

Table 4: One-way ANOVA on per-residue motion amplitude s_1 across 301,979 residues, with grouping by ENSEMBITS primary token and three negative controls. Tokens explain $\sim 7\times$ more variance than the strongest non-token baseline, ruling out fold-class or position-along-chain as alternative explanations of the result in Fig. 8.

Experiments

Flexibility: ANOVA test C_α fluctuation amplitude

s_1 : Local motion amplitude, η^2 : proportion of total variance explained by the token

Tokenizer	M active codes	n residues	η^2 / R^2 on s_1
ENSEMBITS	852	301,979	0.371
Vote_3Di (mini3di plurality, $K=10$)	20	308,293	0.128
ProtProfileMD_K10 (mini3di histogram, R^2)	20	308,293	0.121
AminoAseed	384	307,578	0.079
ESM3struct	1,369	193,613	0.054
3Di_tokens (mini3di, single-frame)	20	308,713	0.051
ProToken	412	304,931	0.006

Table 5: One-way ANOVA on per-residue motion amplitude s_1 (locally Kabsch-aligned $P=10$ C_α frames) across mdCATH-div residues, grouped by primary token from each tokenizer. Active codes M counts only tokens with at least 80 residues assigned. For the continuous ProtProfileMD baseline we report the linear-regression R^2 of $s_1 \sim$ histogram, which is the natural analog of η^2 for continuous predictors. ENSEMBITS captures $\sim 3\times$ the variance of the strongest baselines (Vote_3Di and ProtProfileMD_K10, both multi-frame mini3di aggregations), $\sim 5\times$ that of single-frame AminoAseed, and $\sim 7\times$ that of the single-frame structural tokenizers ESM3struct and 3Di, consistent with its multi-frame, dynamics-aware training objective.

Experiments

Mutation Effect Prediction

ENSEMBITS Setup:

- Uses **ESMFold** to predict the 3D structures for the wild-type (WT) and mutant sequences.
- Static structure passes through the ENSEMBITS tokenizer in its **single-frame mode** ($P=1$).
- For each residue position i , the model calculates the **Euclidean distance** between the WT code embedding and the mutant code embedding
- Distances are summed across all positions in the protein and sum is negated.

Table 2: Zero-shot mutational fitness on PROTEINGYM (96 DMS assays, $L \leq 200$). Per-assay Spearman correlation between predicted score and experimental DMS score, averaged across assays. All tokenizers consume ESMFold-predicted atom14 backbones (real N/C α /C/O/C β); no ideal-geometry reconstruction. The “+ ESM2” rows show the within-assay z -score blend at $\alpha=0.3$ (grid-best for four of five tokenizers); % gain is computed against ESM2 alone. The best result in each block is **bold**, the second-best is underlined.

Model	Mean Spearman	Δ vs ESM2	% gain
3di_tokens (Foldseek/mini3di)	0.327	−0.157	−32.4%
AminoAseed	0.366	−0.118	−24.4%
ProToken	0.358	−0.126	−26.0%
ESM3struct	0.384	−0.100	−20.7%
<u>ENSEMBITS, P=1</u>	<u>0.381</u>	<u>−0.104</u>	<u>−21.4%</u>
ESM2-650M alone	0.484	—	—
+ ESM2 blend ($\alpha=0.3$):			
3di_tokens + ESM2	0.507	+0.023	+4.6%
<u>AminoAseed + ESM2</u>	<u>0.517</u>	<u>+0.033</u>	<u>+6.8%</u>
ProToken + ESM2	0.508	+0.023	+4.8%
ESM3struct + ESM2	0.515	+0.030	+6.3%
ENSEMBITS, P=1 + ESM2	0.518	+0.033	+6.9%

Experiments

Mutation Effect Prediction

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ENSEMBITS, P=1 + ESM2	0.518	+0.033	+6.9%

Experiments

Skill Score

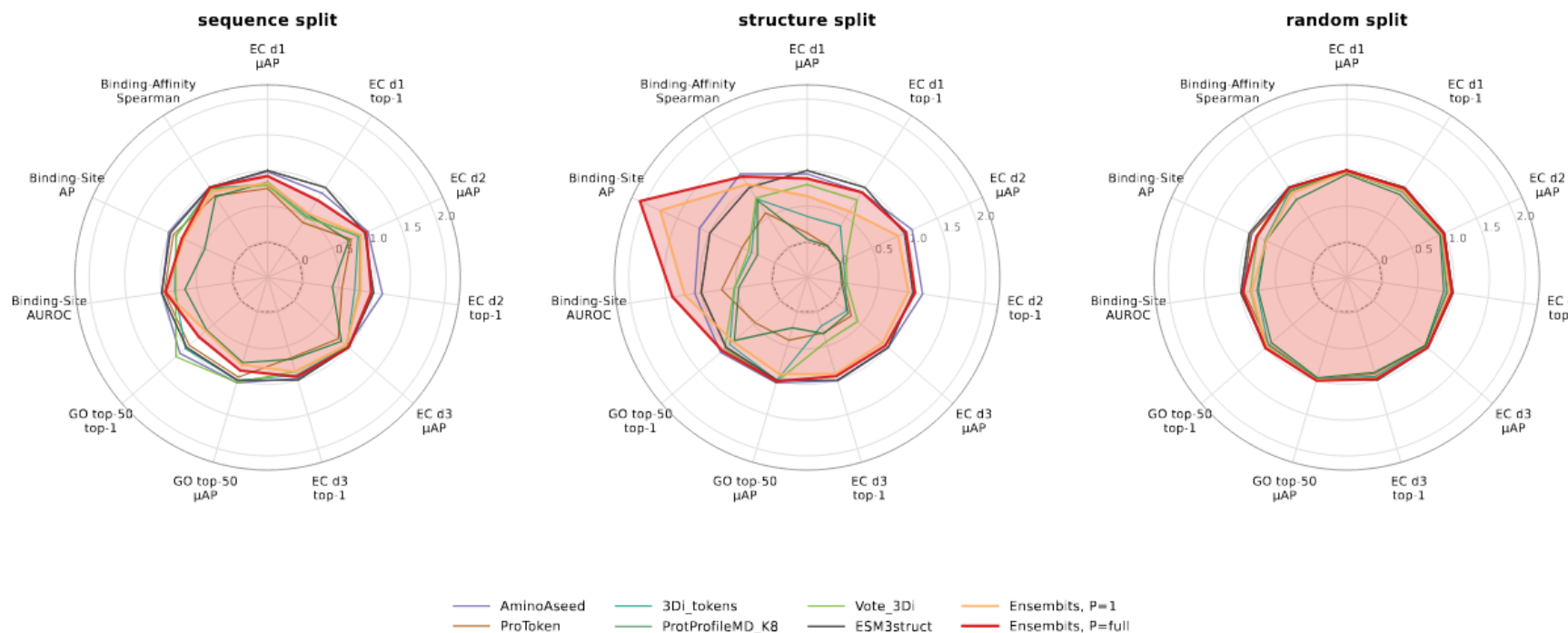


Figure 4: Skill-score radar across MISATO downstream tasks. Each panel is one ProteinShake split; each axis is one (task, metric) pair. Skill is computed per axis and per split as $\text{skill}(m) = (\text{score}(m) - \text{score}(\text{RANDOM})) / (\text{score}(\text{ESM3STRUCT}) - \text{score}(\text{RANDOM}))$, so the inner dashed polygon (0) is the random-token floor and the outer solid polygon (1) is the ESM3struct performance on the same split. Polygons further from the centre recover more of the available functional signal. AA and Random rows are excluded from the polygons for clarity. Seed variance is high.

Experiments

EC Classification

Table 6: EC depth-1 top-1 / mAP / μ AP / F1 — misato structure split, mean $\pm$ std over 10 seeds. Bolding and underlining are separate for the single-frame (top) and multi-frame (bottom) groups.

Model	top-1	mAP	μ AP	F1@0.5	μ F1@0.5
aa	0.333 $\pm$ 0.000	0.346 $\pm$ 0.052	0.350 $\pm$ 0.050	0.080 $\pm$ 0.042	0.266 $\pm$ 0.140
3Di_tokens	0.548 $\pm$ 0.240	0.429 $\pm$ 0.084	0.533 $\pm$ 0.274	0.288 $\pm$ 0.141	0.520 $\pm$ 0.256
Random	0.360 $\pm$ 0.042	0.212 $\pm$ 0.011	0.305 $\pm$ 0.028	0.108 $\pm$ 0.005	0.347 $\pm$ 0.048
AminoAseed	<u>0.841 $\pm$ 0.032</u>	<u>0.569 $\pm$ 0.061</u>	<u>0.912 $\pm$ 0.025</u>	<u>0.514 $\pm$ 0.060</u>	<u>0.830 $\pm$ 0.039</u>
ProToken	<u>0.376 $\pm$ 0.005</u>	<u>0.265 $\pm$ 0.059</u>	<u>0.385 $\pm$ 0.061</u>	<u>0.118 $\pm$ 0.034</u>	<u>0.374 $\pm$ 0.032</u>
ESM3struct	0.882 $\pm$ 0.019	0.615 $\pm$ 0.042	0.943 $\pm$ 0.020	0.548 $\pm$ 0.059	0.869 $\pm$ 0.024
ENSEMBITS, $P = 1$	0.674 $\pm$ 0.239	0.462 $\pm$ 0.115	0.714 $\pm$ 0.260	0.365 $\pm$ 0.157	0.643 $\pm$ 0.284
ProtProfileMD_K8	0.374 $\pm$ 0.000	0.247 $\pm$ 0.043	0.330 $\pm$ 0.039	0.087 $\pm$ 0.046	0.295 $\pm$ 0.155
Vote_3Di	<u>0.774 $\pm$ 0.136</u>	<u>0.504 $\pm$ 0.061</u>	<u>0.818 $\pm$ 0.199</u>	<u>0.422 $\pm$ 0.089</u>	<u>0.765 $\pm$ 0.144</u>
ENSEMBITS, $P = full$	0.839 $\pm$ 0.033	0.543 $\pm$ 0.045	0.871 $\pm$ 0.045	0.456 $\pm$ 0.082	0.800 $\pm$ 0.125

Experiments

GO Classification

Table 7: GO top-50 — misato structure split, mean $\pm$ std over 10 seeds. Bolding and underlining are separate for the single-frame (top) and multi-frame (bottom) groups.

Model	top-1	mAP	μ AP	F1@0.5	μ F1@0.5
aa	0.354 $\pm$ 0.348	0.300 $\pm$ 0.141	0.500 $\pm$ 0.301	0.116 $\pm$ 0.186	0.299 $\pm$ 0.411
3Di_tokens	0.641 $\pm$ 0.052	0.400 $\pm$ 0.021	<u>0.837 $\pm$ 0.026</u>	0.304 $\pm$ 0.040	<u>0.777 $\pm$ 0.072</u>
Random	0.294 $\pm$ 0.163	0.114 $\pm$ 0.005	<u>0.247 $\pm$ 0.058</u>	0.000 $\pm$ 0.000	<u>0.000 $\pm$ 0.001</u>
AminoAseed	0.704 $\pm$ 0.048	0.444 $\pm$ 0.021	0.859 $\pm$ 0.018	0.374 $\pm$ 0.028	0.814 $\pm$ 0.022
ProToken	0.468 $\pm$ 0.224	0.208 $\pm$ 0.108	0.492 $\pm$ 0.243	0.084 $\pm$ 0.127	0.257 $\pm$ 0.359
ESM3struct	<u>0.665 $\pm$ 0.073</u>	<u>0.428 $\pm$ 0.040</u>	0.835 $\pm$ 0.101	<u>0.342 $\pm$ 0.083</u>	0.747 $\pm$ 0.220
ENSEMBITS, P=1	<u>0.619 $\pm$ 0.057</u>	<u>0.397 $\pm$ 0.033</u>	0.781 $\pm$ 0.071	<u>0.283 $\pm$ 0.073</u>	0.681 $\pm$ 0.127
ProtProfileMD_K8	0.610 $\pm$ 0.073	0.199 $\pm$ 0.074	0.386 $\pm$ 0.166	0.040 $\pm$ 0.108	0.087 $\pm$ 0.252
Vote_3Di	<u>0.676 $\pm$ 0.073</u>	<u>0.423 $\pm$ 0.020</u>	0.849 $\pm$ 0.030	<u>0.335 $\pm$ 0.038</u>	<u>0.790 $\pm$ 0.043</u>
ENSEMBITS, P=full	0.692 $\pm$ 0.073	0.428 $\pm$ 0.033	<u>0.846 $\pm$ 0.047</u>	0.347 $\pm$ 0.052	0.798 $\pm$ 0.065

Experiments

Binding Site

Table 8: Binding-site — misato structure split, mean $\pm$ std over 10 seeds. Bolding and underlining are separate for the single-frame (top) and multi-frame (bottom) groups.

Model	AUROC	AP
aa	0.649 $\pm$ 0.054	0.300 $\pm$ 0.095
3Di_tokens	0.594 $\pm$ 0.012	0.184 $\pm$ 0.020
Random	0.500 $\pm$ 0.005	0.124 $\pm$ 0.002
AminoAseed	<u>0.694 $\pm$ 0.032</u>	<u>0.318 $\pm$ 0.067</u>
ProToken	<u>0.626 $\pm$ 0.007</u>	<u>0.208 $\pm$ 0.013</u>
ESM3struct	0.678 $\pm$ 0.043	0.291 $\pm$ 0.110
ENSEMBITS, P=1	0.719 $\pm$ 0.019	0.418 $\pm$ 0.032
ProtProfileMD_K8	0.583 $\pm$ 0.004	0.169 $\pm$ 0.005
Vote_3Di	<u>0.594 $\pm$ 0.015</u>	<u>0.191 $\pm$ 0.025</u>
ENSEMBITS, P=full	0.750 $\pm$ 0.011	0.471 $\pm$ 0.010

Experiments

Binding Affinity

Table 9: Binding affinity ($-\log K_d/K_i$) — misato structure split, ligand-aware head (MACCS-167), mean $\pm$ std over 10 seeds. Bolding and underlining are separate for the single-frame (top) and multi-frame (bottom) groups. **MSE: lower is better.**

Model	R^2	Spearman	MSE
aa	0.113 $\pm$ 0.082	0.411 $\pm$ 0.034	2.862 $\pm$ 0.265
3Di_tokens	0.080 $\pm$ 0.063	0.378 $\pm$ 0.041	3.000 $\pm$ 0.207
Random	<u>0.103 $\pm$ 0.053</u>	0.389 $\pm$ 0.032	<u>2.924 $\pm$ 0.173</u>
AminoAseed	<u>0.039 $\pm$ 0.126</u>	0.465 $\pm$ 0.034	<u>3.133 $\pm$ 0.411</u>
ProToken	0.091 $\pm$ 0.052	0.376 $\pm$ 0.034	2.965 $\pm$ 0.169
ESM3struct	0.077 $\pm$ 0.074	<u>0.432 $\pm$ 0.060</u>	3.009 $\pm$ 0.240
ENSEMBITS, P=1	0.092 $\pm$ 0.082	<u>0.412 $\pm$ 0.045</u>	2.959 $\pm$ 0.267
ProtProfileMD_K8	0.058 $\pm$ 0.056	0.364 $\pm$ 0.042	3.070 $\pm$ 0.183
Vote_3Di	<u>0.075 $\pm$ 0.060</u>	<u>0.400 $\pm$ 0.045</u>	<u>3.016 $\pm$ 0.194</u>
ENSEMBITS, P=full	0.090 $\pm$ 0.082	0.415 $\pm$ 0.052	2.966 $\pm$ 0.267

Experiments

Codebook Utilization

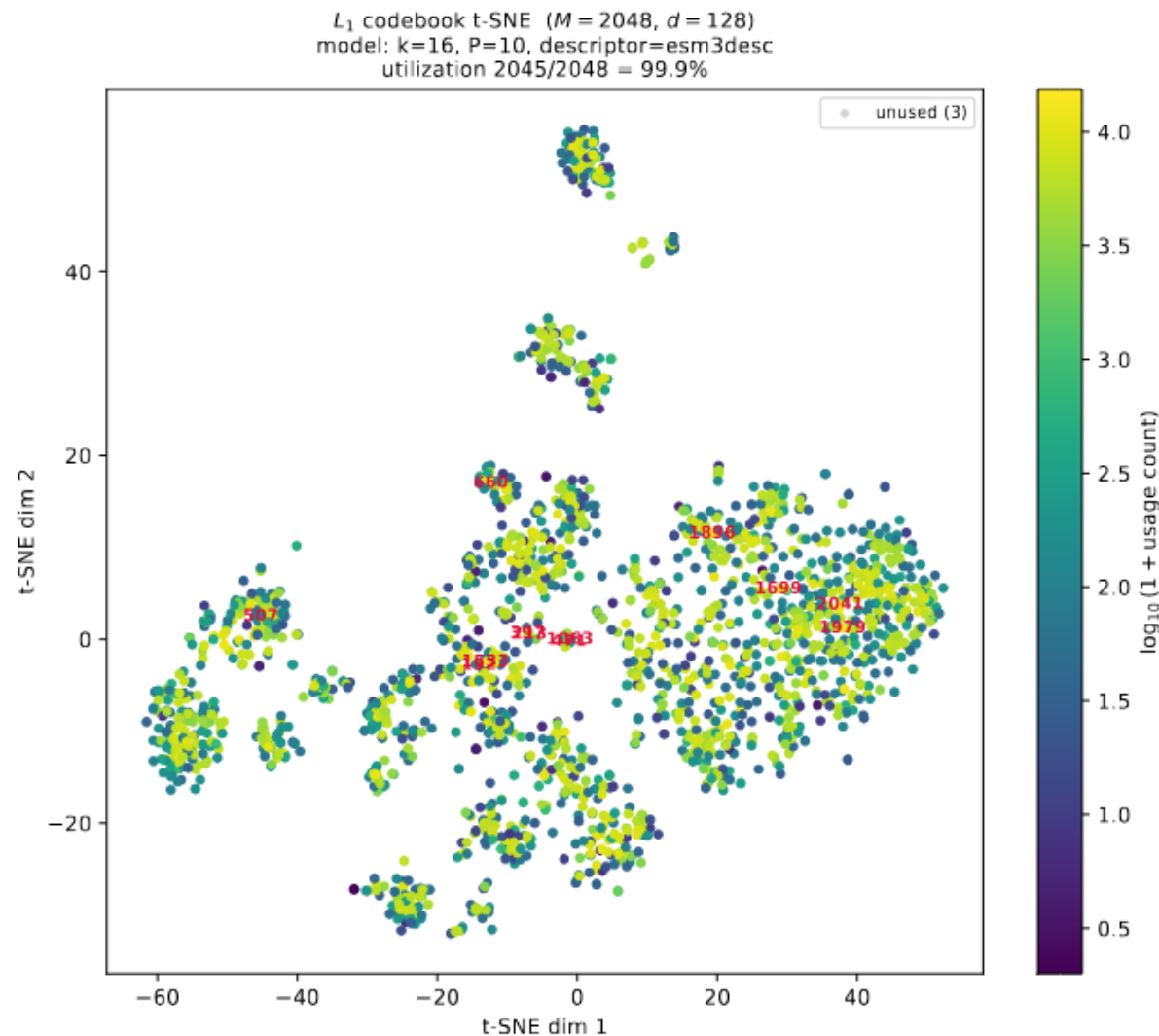


Figure 7: t-SNE projection of the L_1 primary codebook ($M = 2048$ entries, $d = 128$) for the production tokenizer (combined mdCATH + MISATO training, $k = 16$, $P = 10$). Each point is one codebook entry; the color of used codes encodes $\log_{10}(1 + \text{usage count})$ across the combined corpus, while the 3 unused codes are drawn in light gray. The twelve most-used token IDs are annotated in red. Codebook utilization is 99.85% with a near-uniform usage distribution (top-to-mean ratio $\approx 4.3\times$), and the t-SNE shows no separation between high- and low-usage codes — the codebook tiles the latent manifold approximately uniformly.

Discussion

Things that **didn't** work

- **3Di-style Descriptors:** ESM3-style relative-frame descriptors performed empirically better and provided an information upper bound over the 3Di construction.
- **Single-Codebook VQ-VAE:** Using a standard VQ-VAE with a single codebook was found to be less effective. The authors noted that the subsequent levels in the RVQ allow the first level to focus on a "denoised" version of the dominant dynamical mode.
- **Gradient-Based Codebook Updates:** Standard gradient-driven codebook losses (as opposed to EMA) led to slow, gradient-driven oscillations during training.
- **Fixed and Fused Neighbor Selection Modes:**
 - The **Fixed** mode: Neighbor selection at a single reference frame, cannot capture the formation and breakage of contacts along a trajectory.
 - The **Fused** mode: Concatenates all neighbors appearing in any frame, results in a P-dependent dimension that does not permit Single-Frame-to-Token Distillation (SFTD)
- **Training Without Cosine Annealing:** Without a cosine learning rate schedule, late-stage updates were large enough to perturb a converged codebook and cause token utilization to oscillate.
- **Geometry-Only Features for Binding Affinity:** On the challenging "structure split" for binding-affinity regression, the researchers found that geometry-only single-frame tokenizers (including ENSEMBITS at P=1) provided no measurable signal over a simple amino-acid (aa) baseline.

Thanks...